CSL-351: Computer Networks

Lab - 10

Cisco Packet Tracer

Dr. Anand M. Baswade
Indian Institute of Technology, Bhilai
anand@iitbhilai.ac.in

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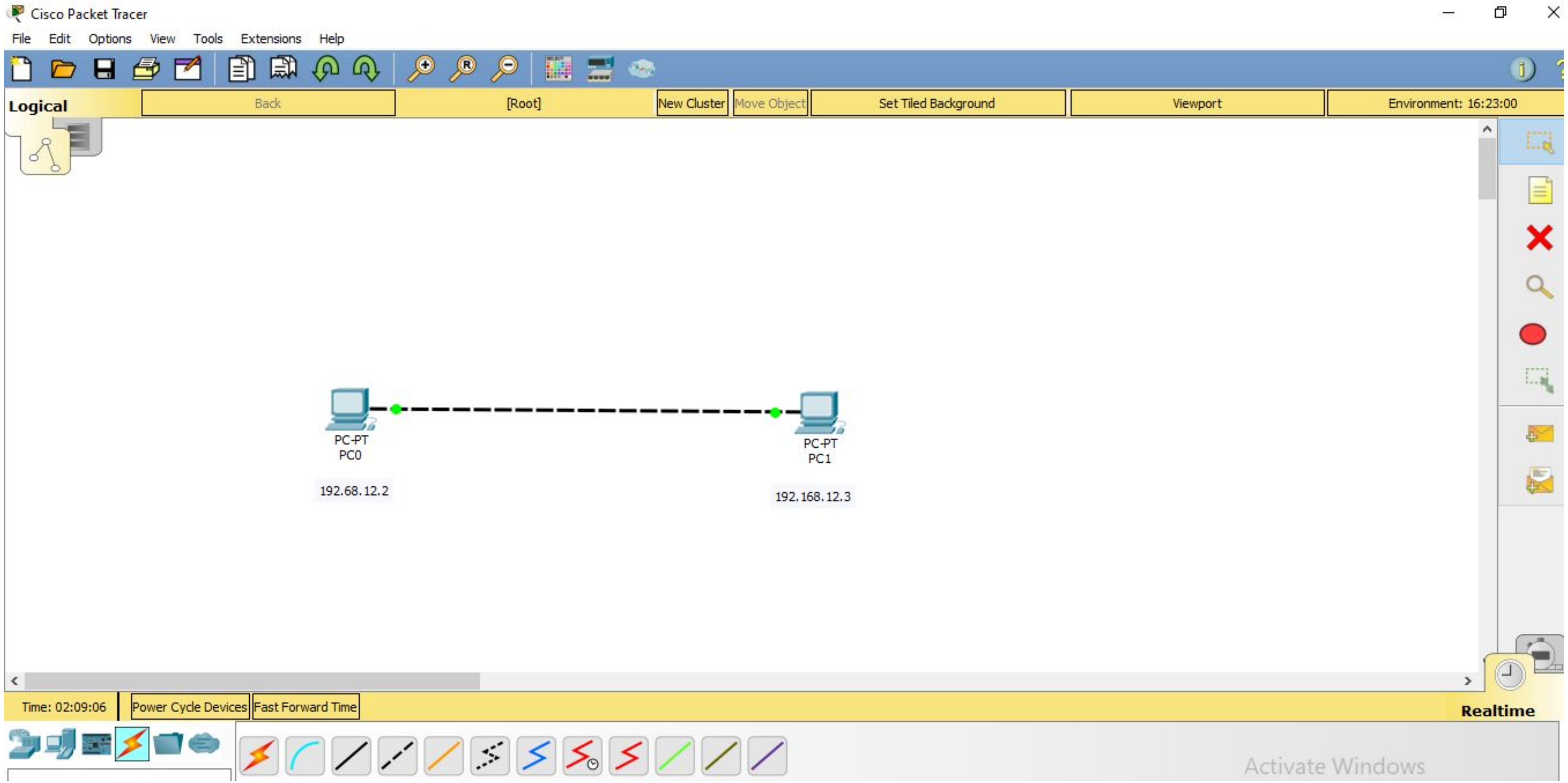
- ❑ Installation
- ❑ Assign IP’s in CPT
- ❑ Connection with Hub
- ❑ Connection with Switch
- ❑ Creation of Multiple LAN’s
- ❑ Creation of WAN’s
- ❑ DHCP

Introduction

- Cisco Packet Tracer is a network simulation software developed by Cisco Systems
- **Learning network concepts:** Packet Tracer can help you learn about the fundamentals of networking, such as the OSI model, TCP/IP, subnetting, and routing.
- **Designing networks:** You can use Packet Tracer to design new networks or to improve existing networks.
- **Troubleshooting networks:** Packet Tracer can help you identify and troubleshoot network problems in a safe virtual environment.

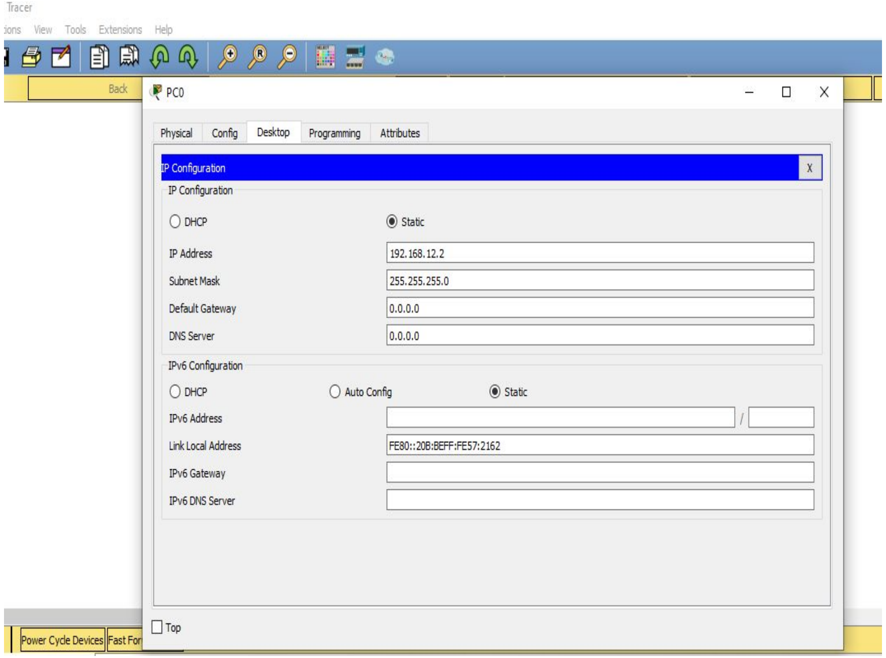
How to level the IP address



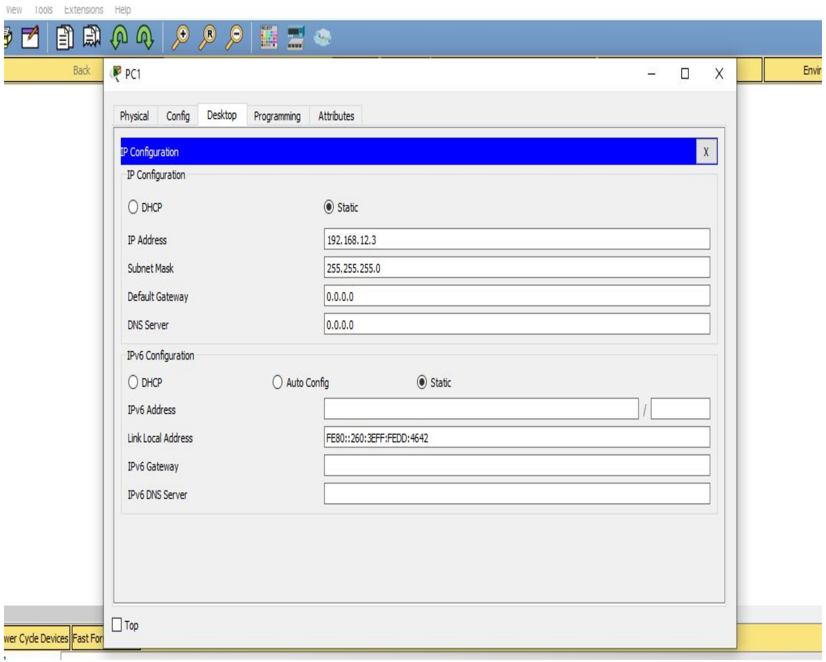
Steps

- Allocate IP addresses to the PCs

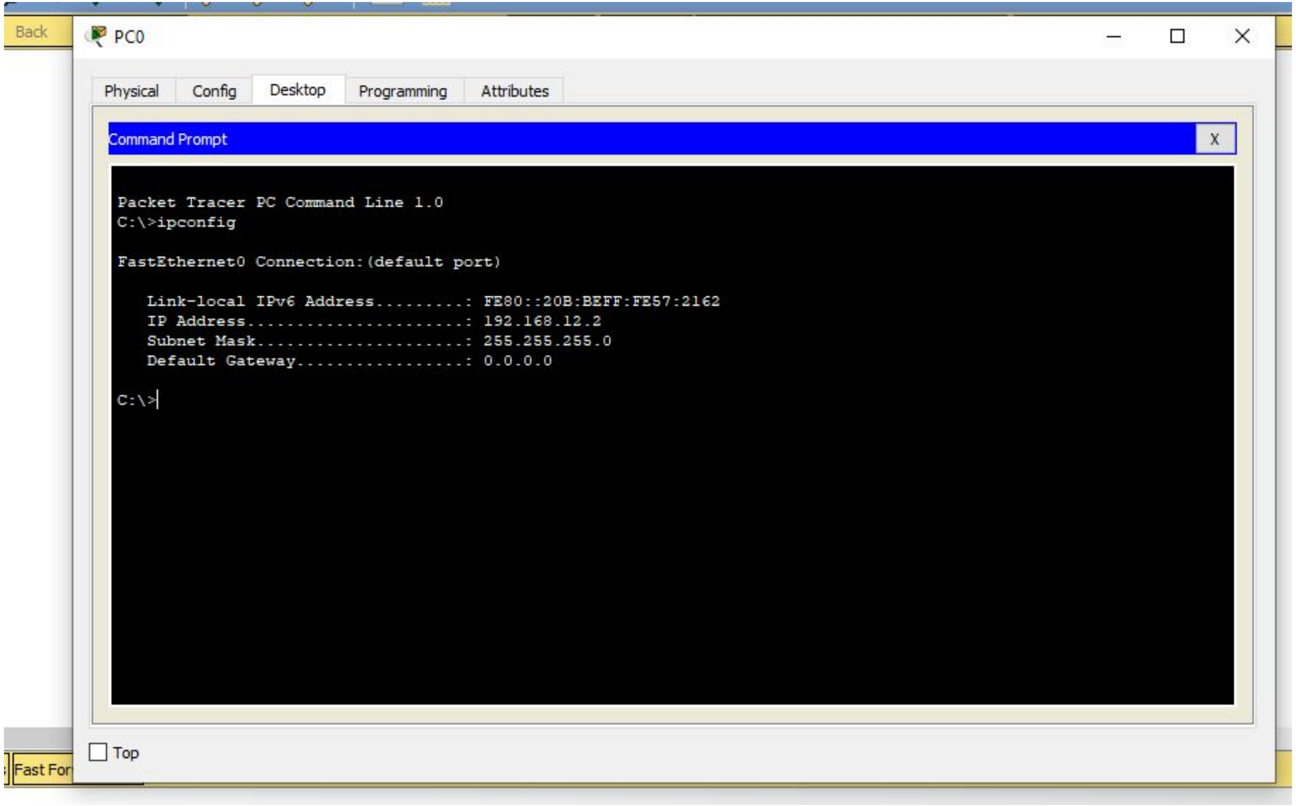
PC0



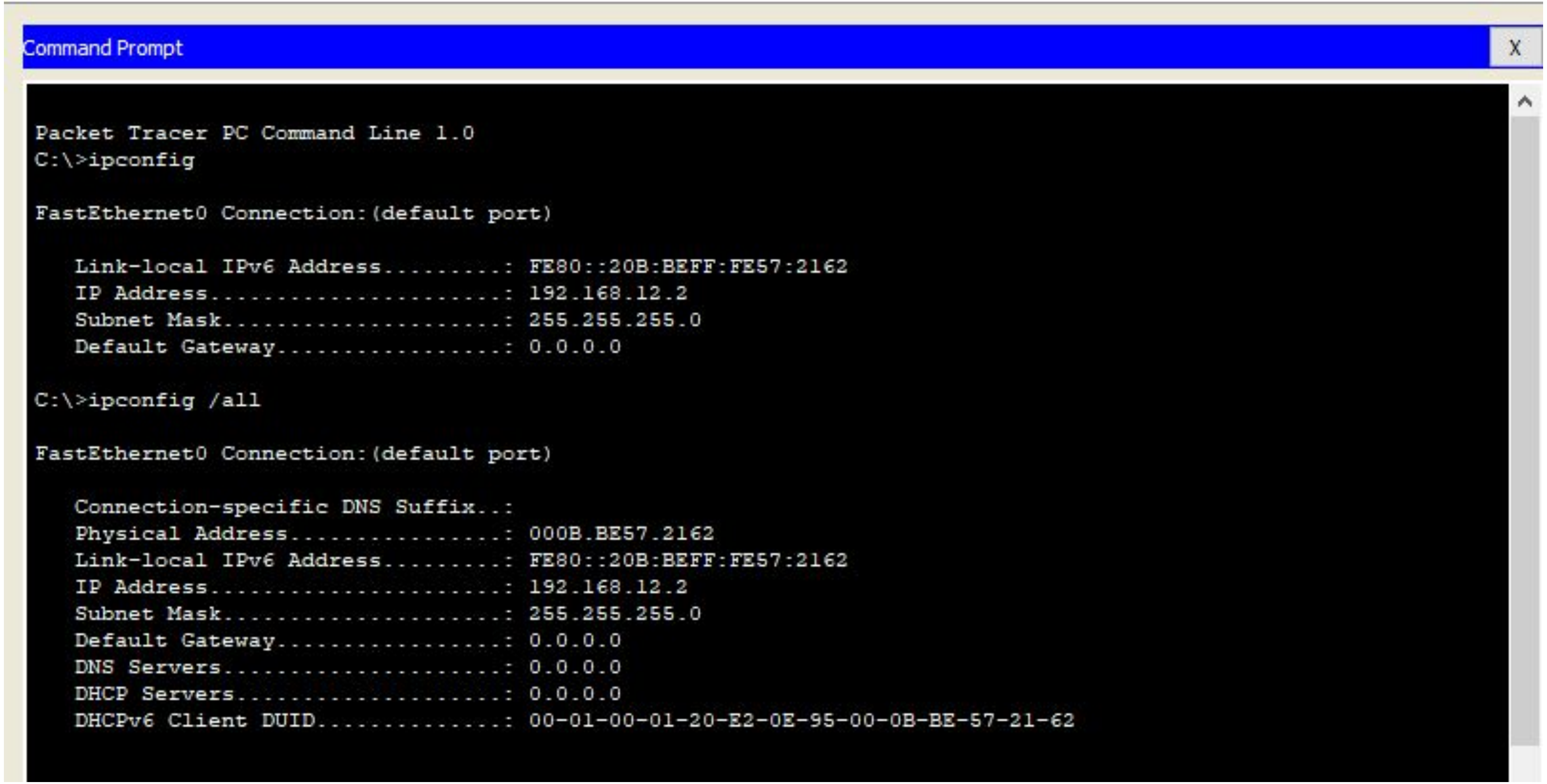
PC1



To check the ip address of PC0

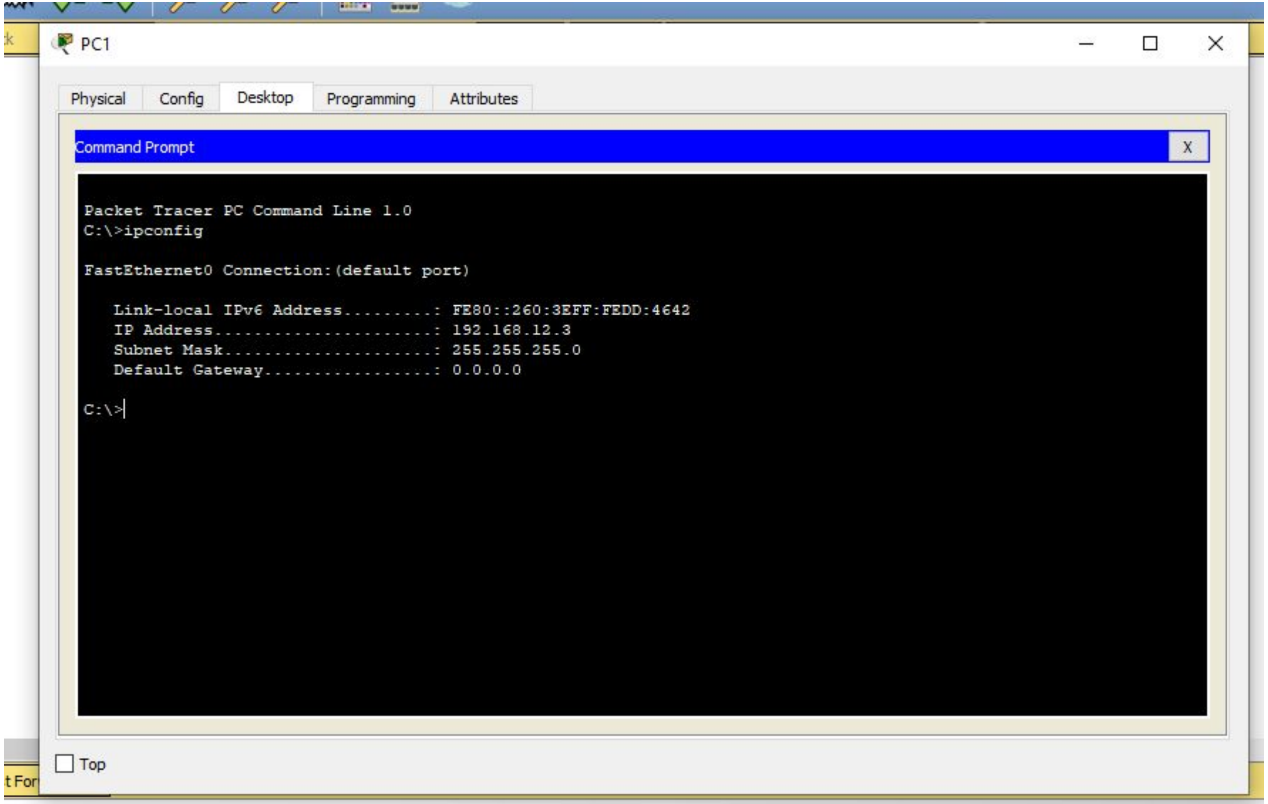


To check the MAC address

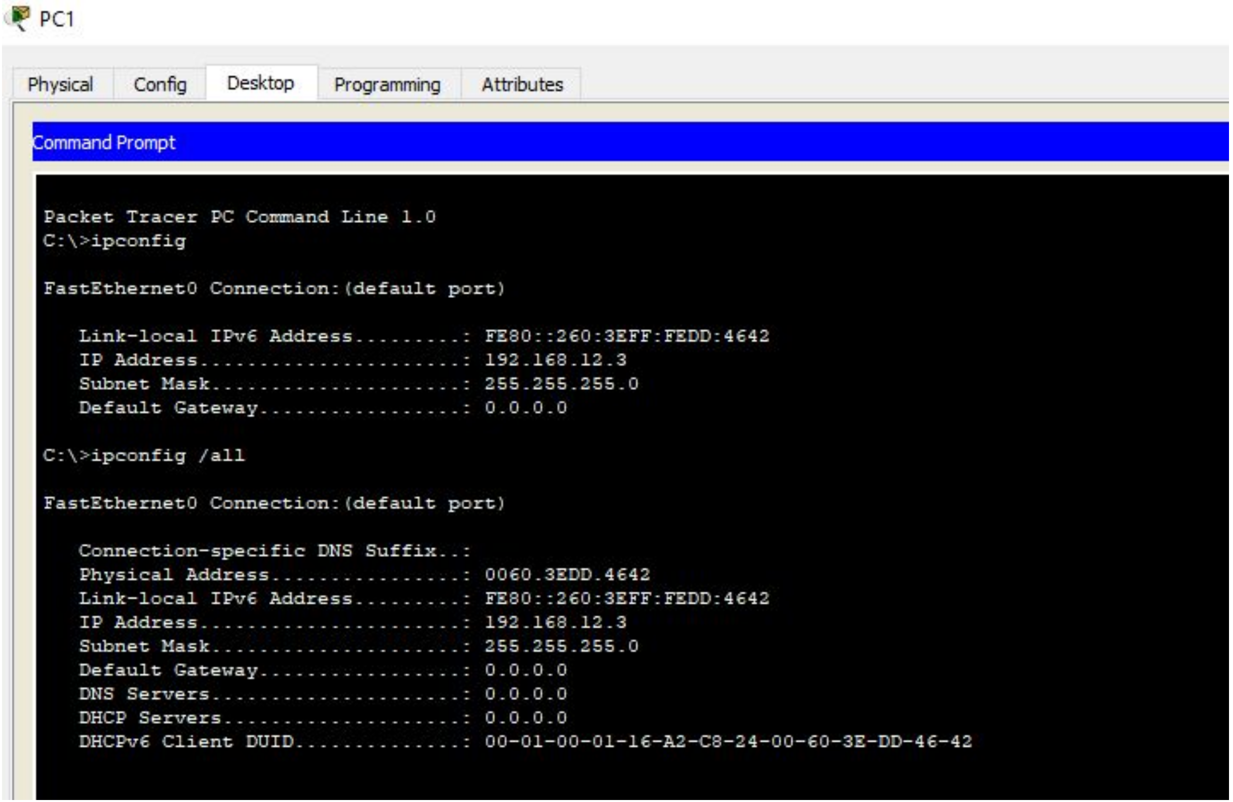


```
Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Link-local IPv6 Address.........: FE80::20B:BEFF:FE57:2162
   IP Address......................: 192.168.12.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 0.0.0.0

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 000B.BE57.2162
   Link-local IPv6 Address.........: FE80::20B:BEFF:FE57:2162
   IP Address......................: 192.168.12.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 0.0.0.0
   DNS Servers.....................: 0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 Client DUID..............: 00-01-00-01-20-E2-0E-95-00-0B-BE-57-21-62
```

To check the ip address of PC1



To check the MAC address of PC1



Connection of switch to form LAN

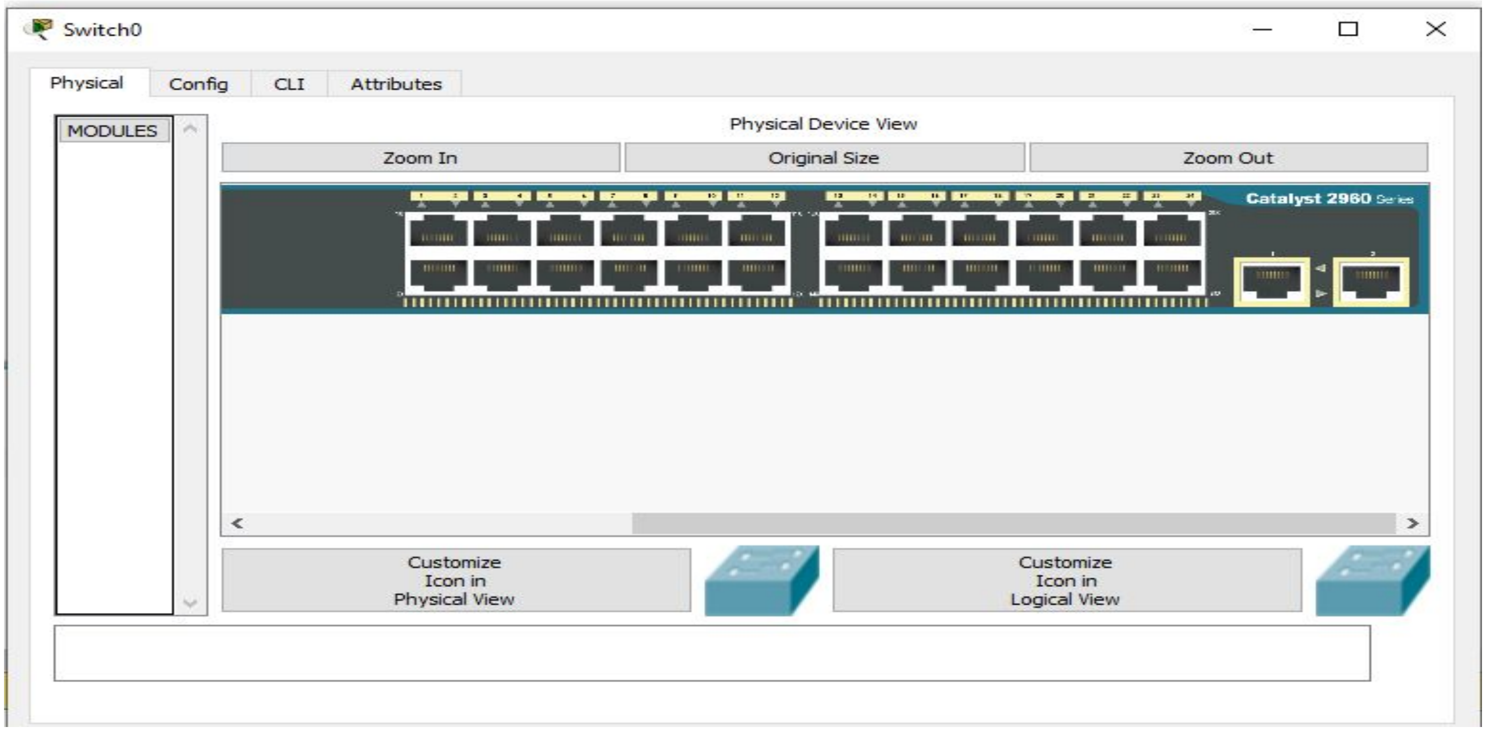
- Connect End Devices to the ports of Switch
- Use copper straight through cable to connect
- Give ip addresses to all the individual pc, level the ip address for each pc
- Check using ping command
- Go to simulation send the packet

Connection of switch

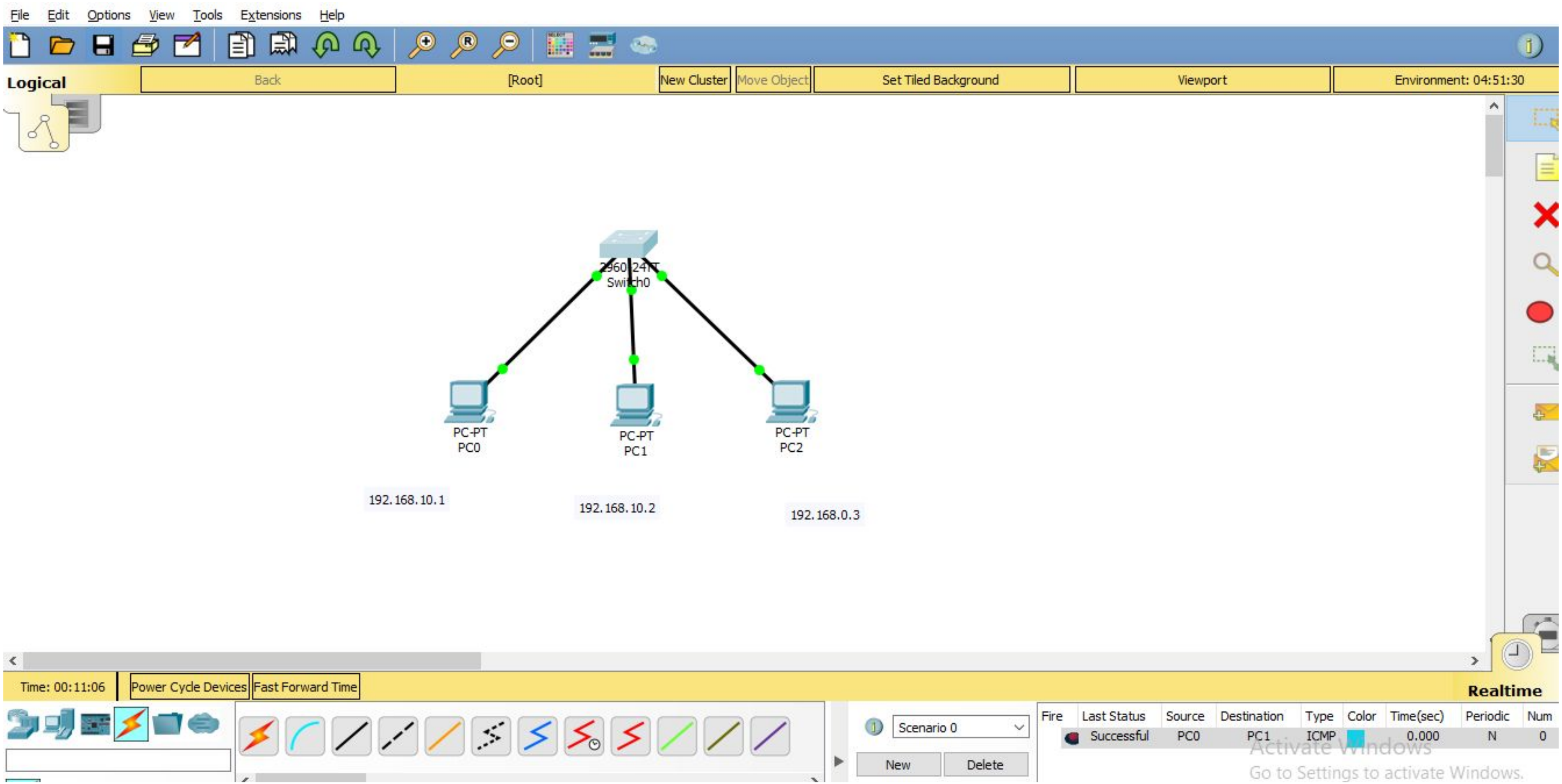
Creation of LAN using switch

- Create single local area network
- Required components
  - 3 end devices
  - 1 switch

Physical view of switch

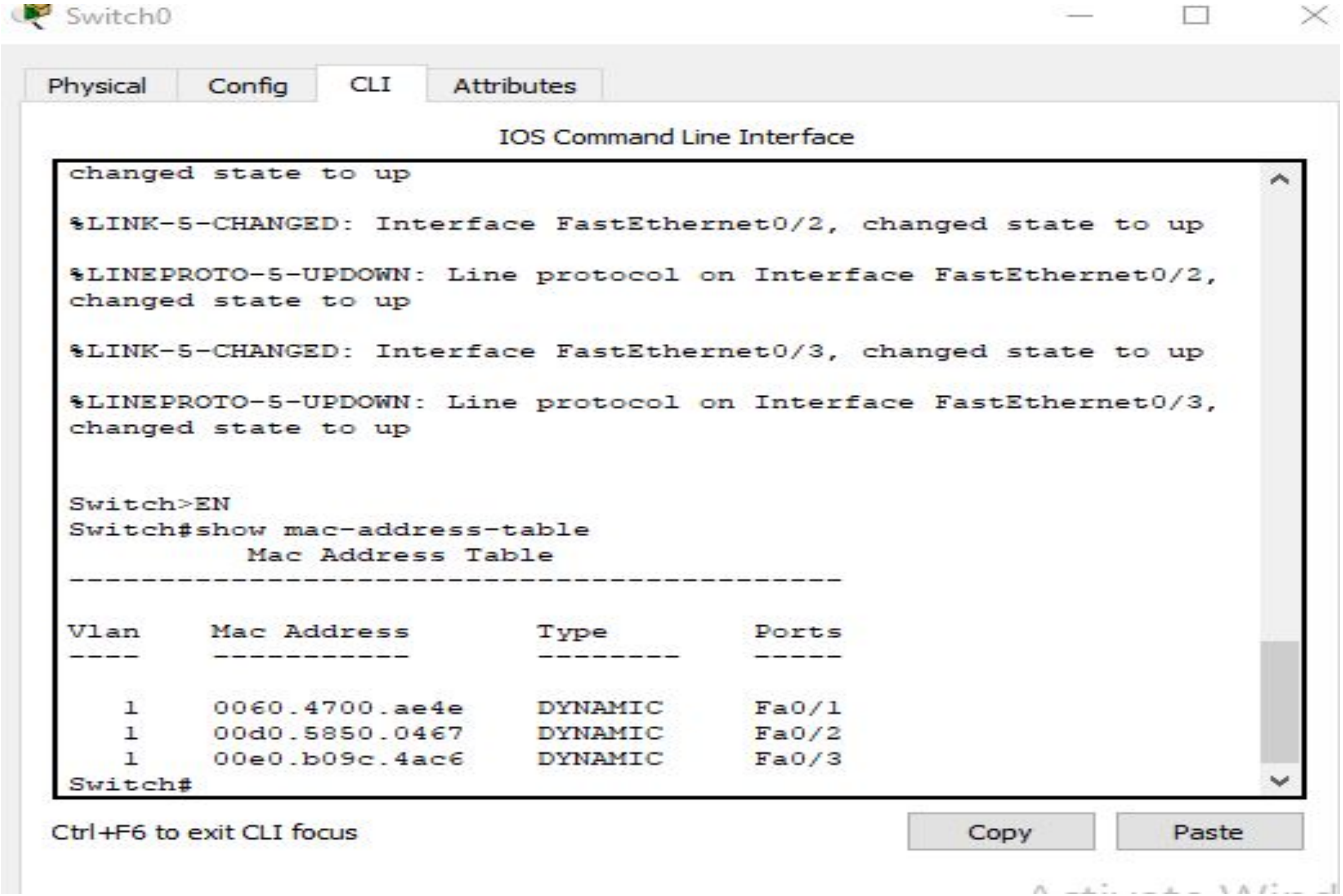


24 Fast Ethernet port and 2 Gigabit Ethernet port


File Edit Options View Tools Extensions Help
Logical
Back
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Environment: 04:51:30
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
192.168.10.1
192.168.10.2
192.168.0.3
Time: 00:11:06
Power Cycle Devices
Fast Forward Time
Realtime
Scenario 0
New
Delete
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Successful
PC0
PC1
ICMP
0.000
N
0
Activate Windows
Go to Settings to activate Windows.

MAC Address Tables

- A MAC address table, sometimes called a Content Addressable Memory (CAM) table, is used on Ethernet switches to determine where to forward traffic on a LAN.
- Now let's break this down a little bit to understand how the MAC address table is built and used by an Ethernet switch to help traffic move along the path to its destination.


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
Switch>EN
Switch#show mac-address-table
Mac Address Table

Vlan Mac Address Type Ports
---- ----------- -------- -----
1 0060.4700.ae4e DYNAMIC Fa0/1
1 00d0.5850.0467 DYNAMIC Fa0/2
1 00e0.b09c.4ac6 DYNAMIC Fa0/3
Switch#
Ctrl+F6 to exit CLI focus
Copy
Paste

Creation of Multiple LAN

- Create two local area network and connect via router
- Required components
  - 6 end devices
  - 2 switches (2960)
  - 1 router (2911)

Steps

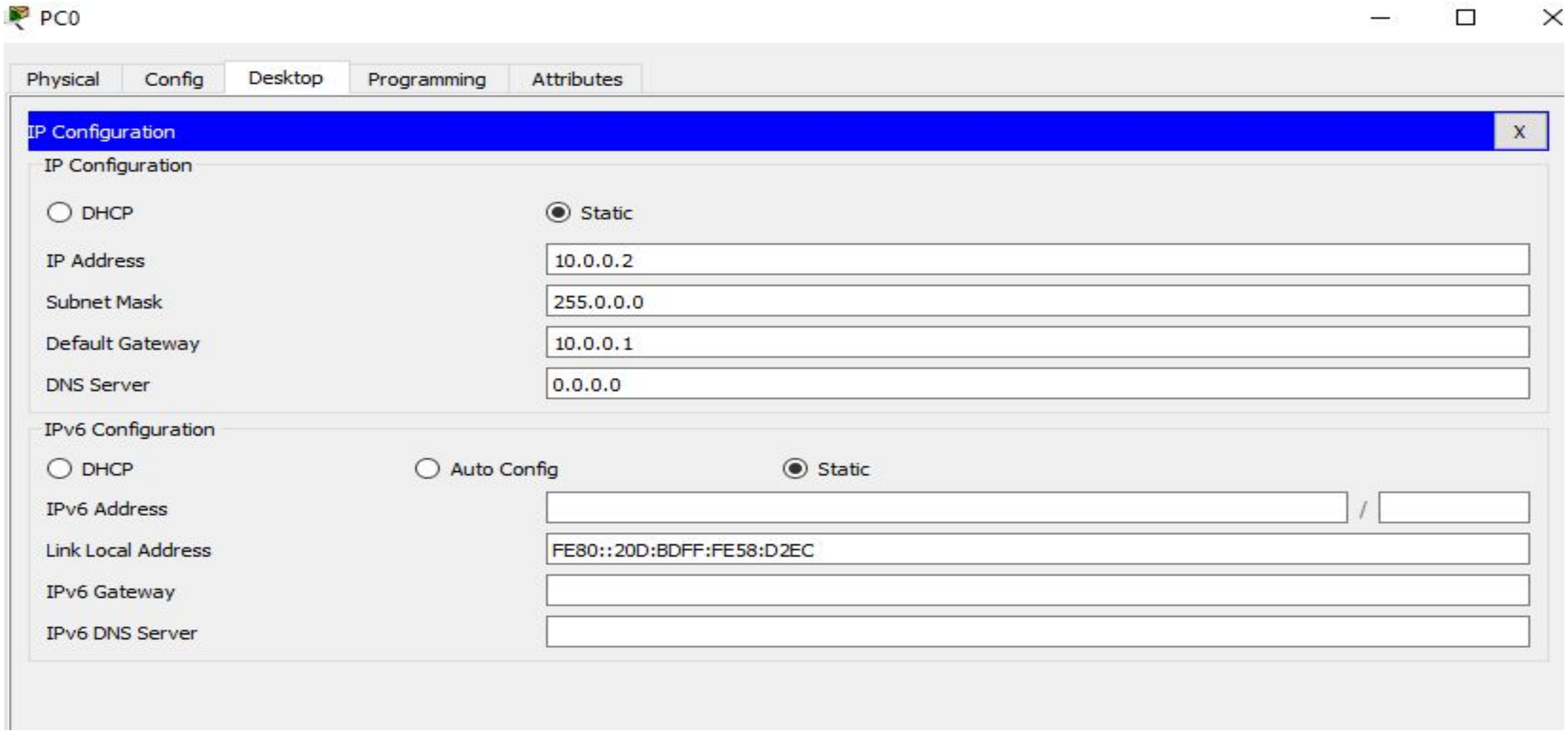
- Place all the devices in the workspace
- Assign IP addresses to all the PCs of LAN1

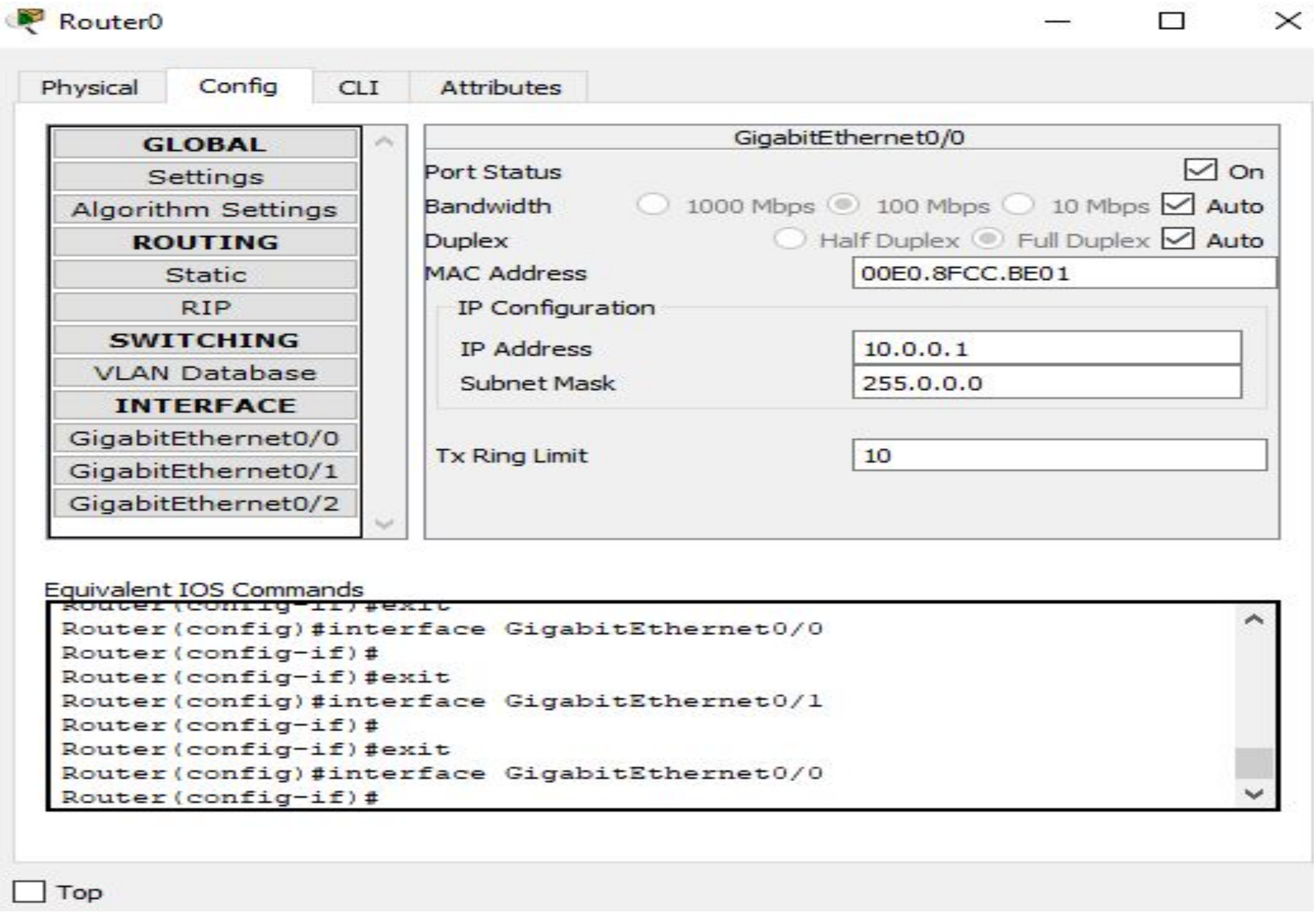
    10.0.0.2, 10.0.0.3, 10.0.0.4

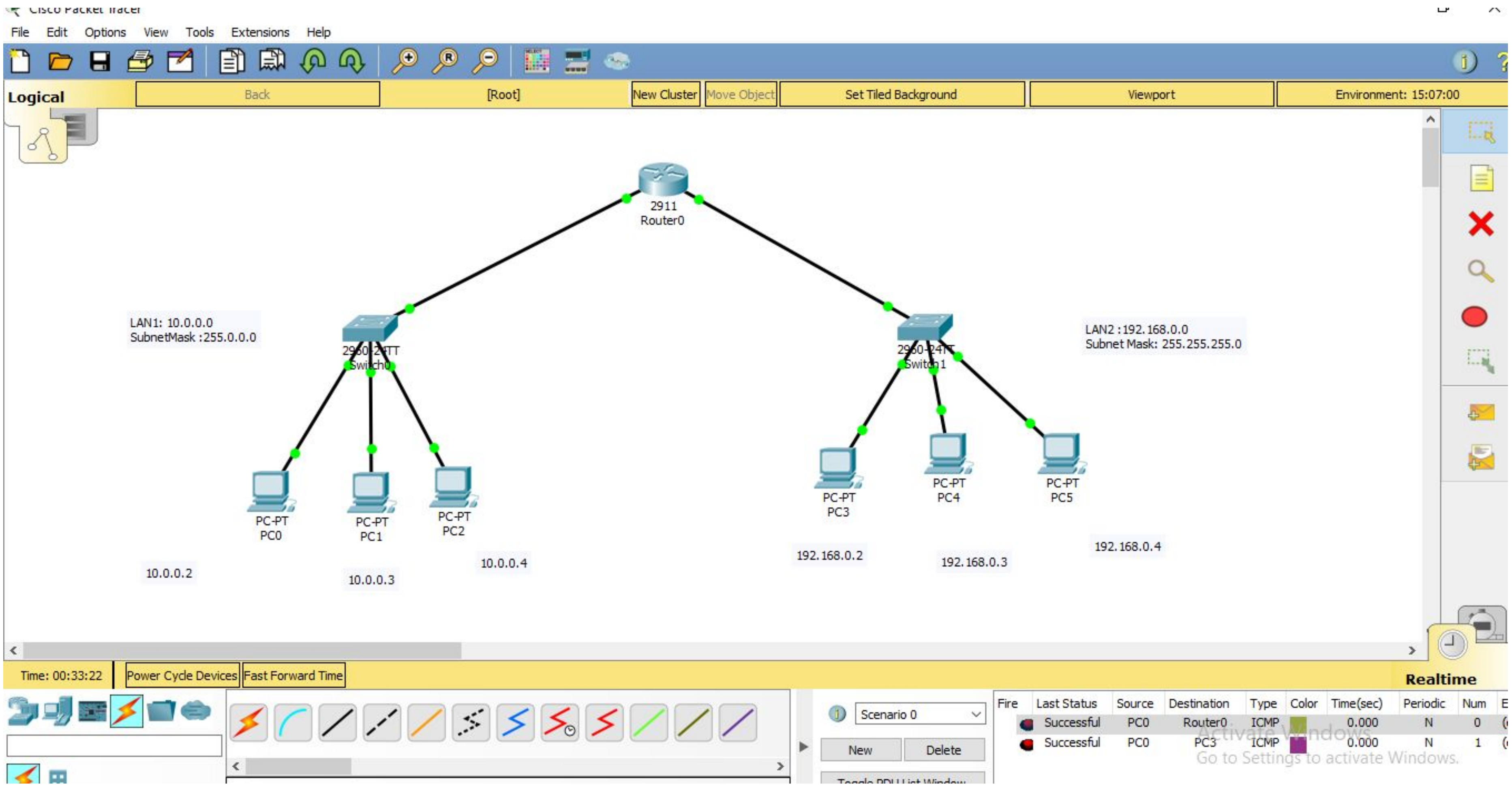
- Assign IP addresses to all the PCs of LAN2

    192.168.0.2, 192.168.0.3, 192.168.0.4

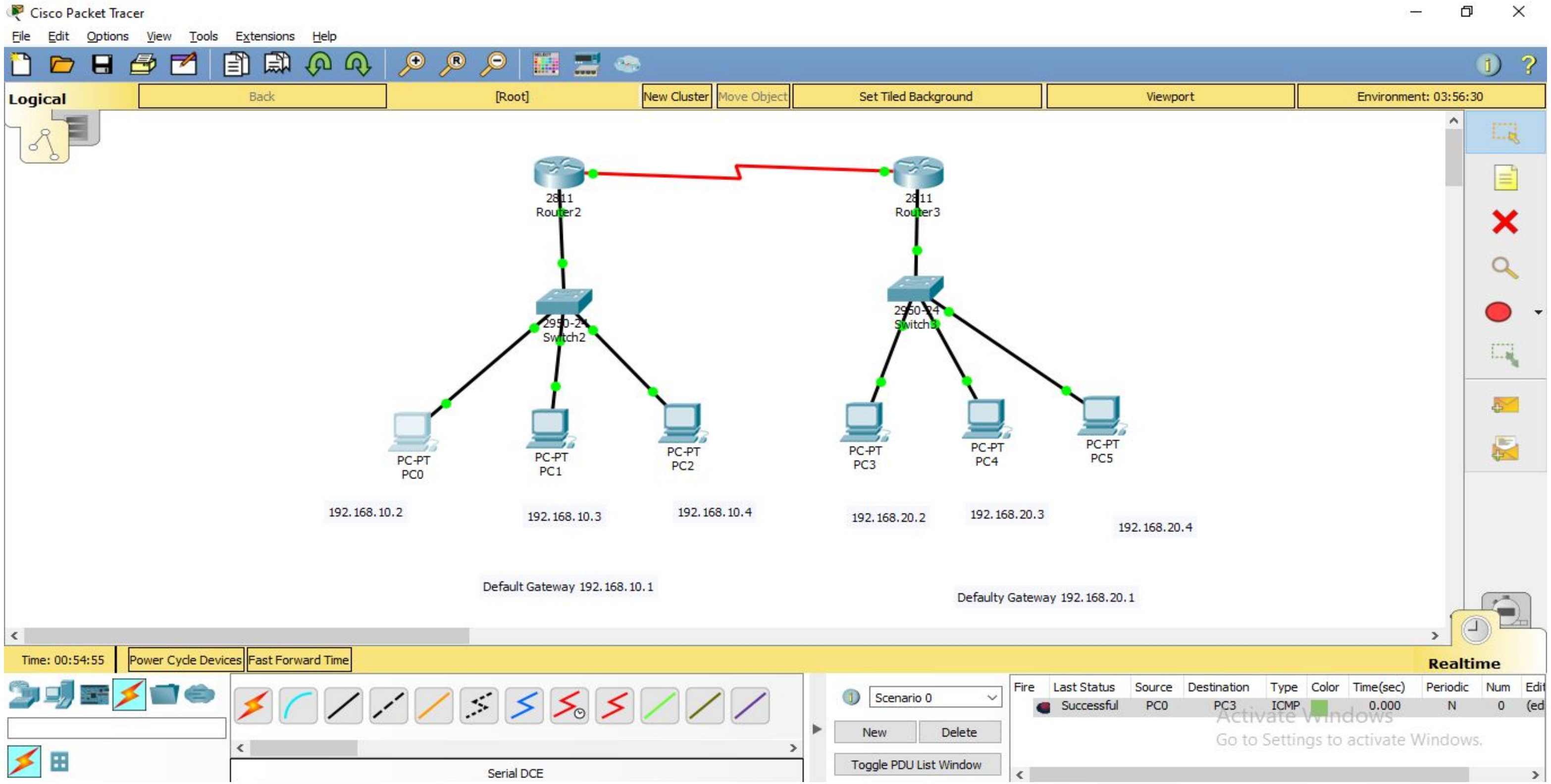
Set the default gateway for each end device




Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.8FCC.BE01
IP Configuration
IP Address
10.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10
Equivalent IOS Commands
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
Back
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Environment: 15:07:00
2911
Router0
LAN1: 10.0.0.0
SubnetMask :255.0.0.0
2960-24TT
Switch0
2960-24TT
Switch1
LAN2 :192.168.0.0
Subnet Mask: 255.255.255.0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
10.0.0.2
10.0.0.3
10.0.0.4
192.168.0.2
192.168.0.3
192.168.0.4
Time: 00:33:22
Power Cycle Devices
Fast Forward Time
Realtime
Scenario 0
New
Delete
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Successful PC0 Router0 ICMP 0.000 N 0
Successful PC0 PC3 ICMP 0.000 N 1
Activate Windows
Go to Settings to activate Windows.

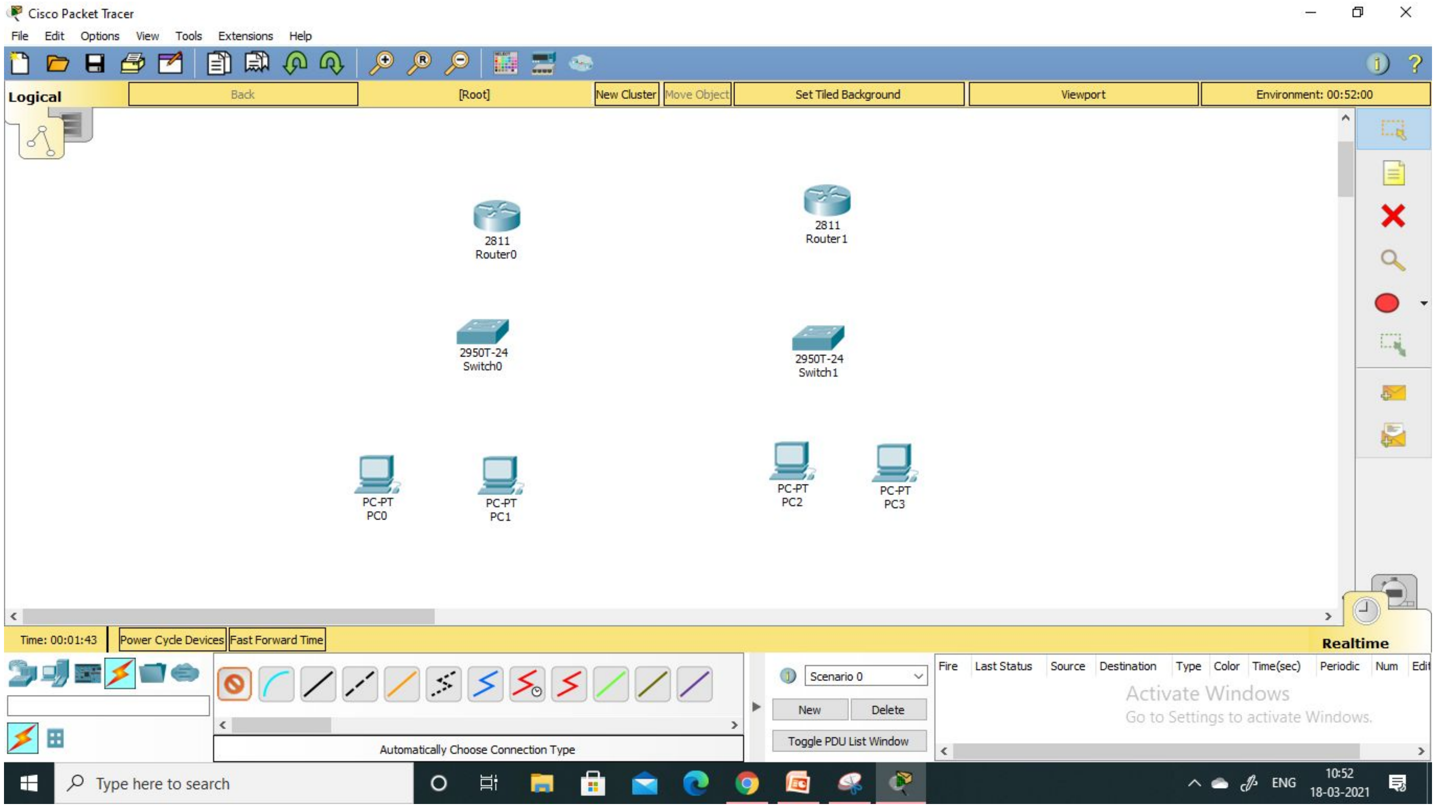
Creation of WAN

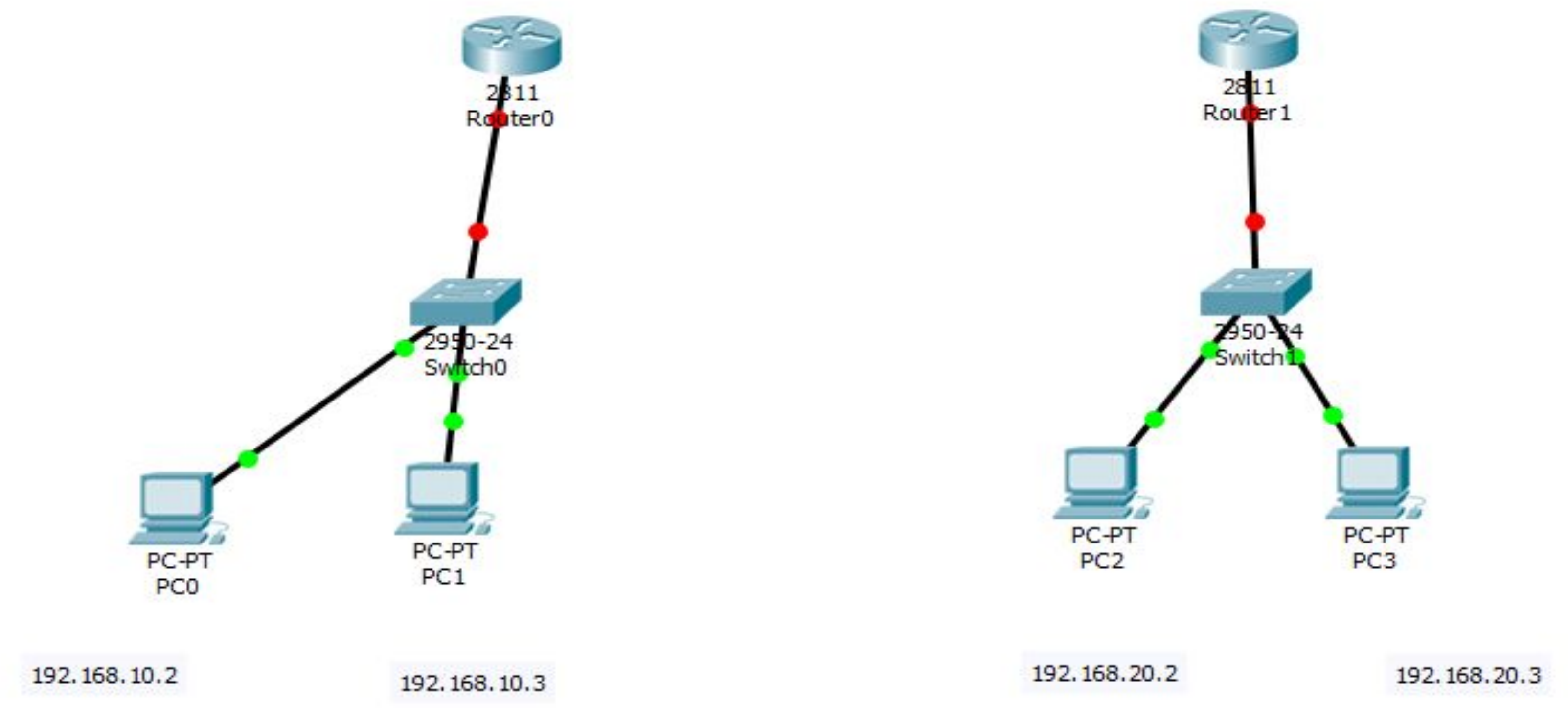


Creation of Network

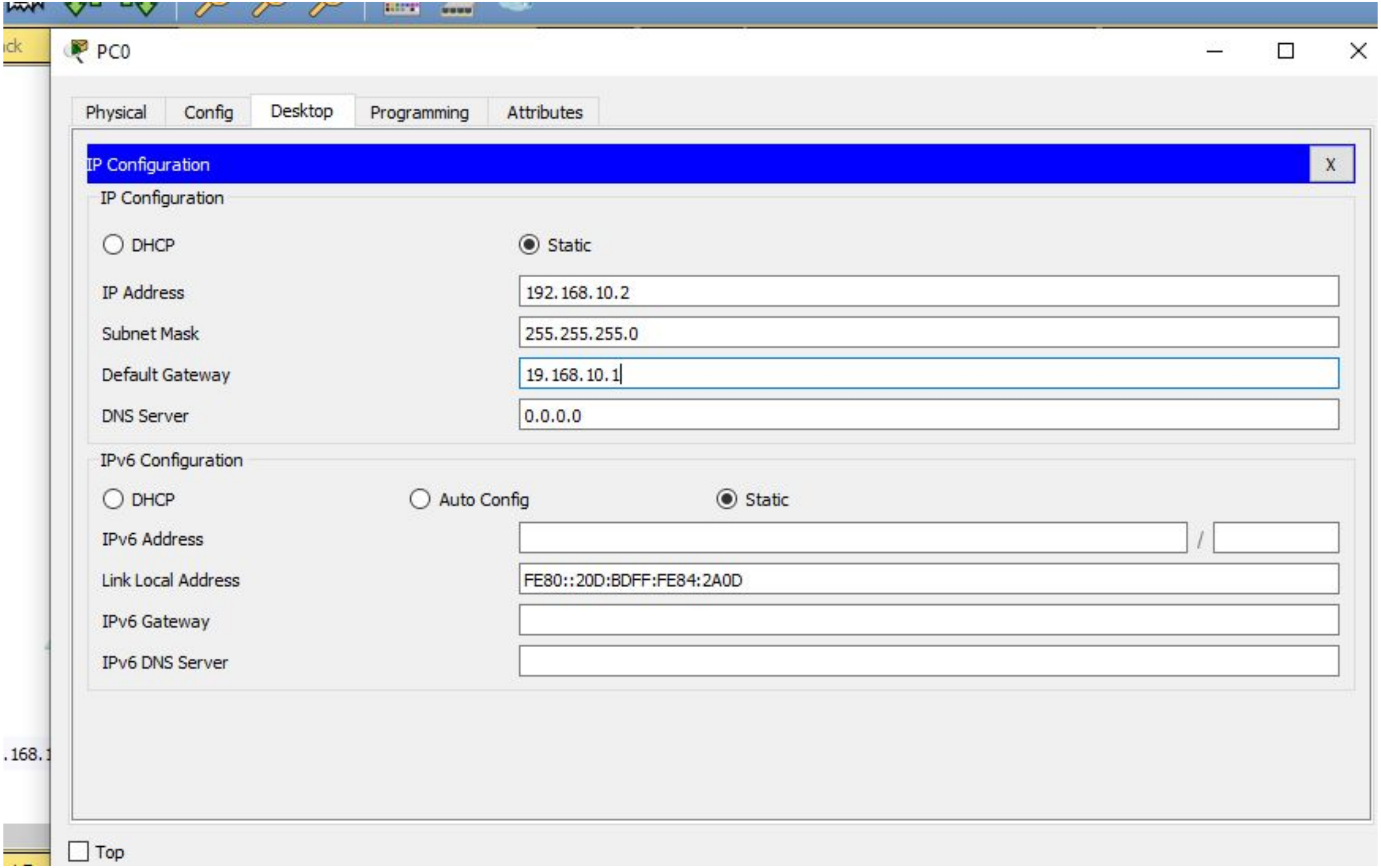
- Create single network
- Required components
  - 6 end devices
  - 2 switch (2950-24T)
  - 2 Router (2811)

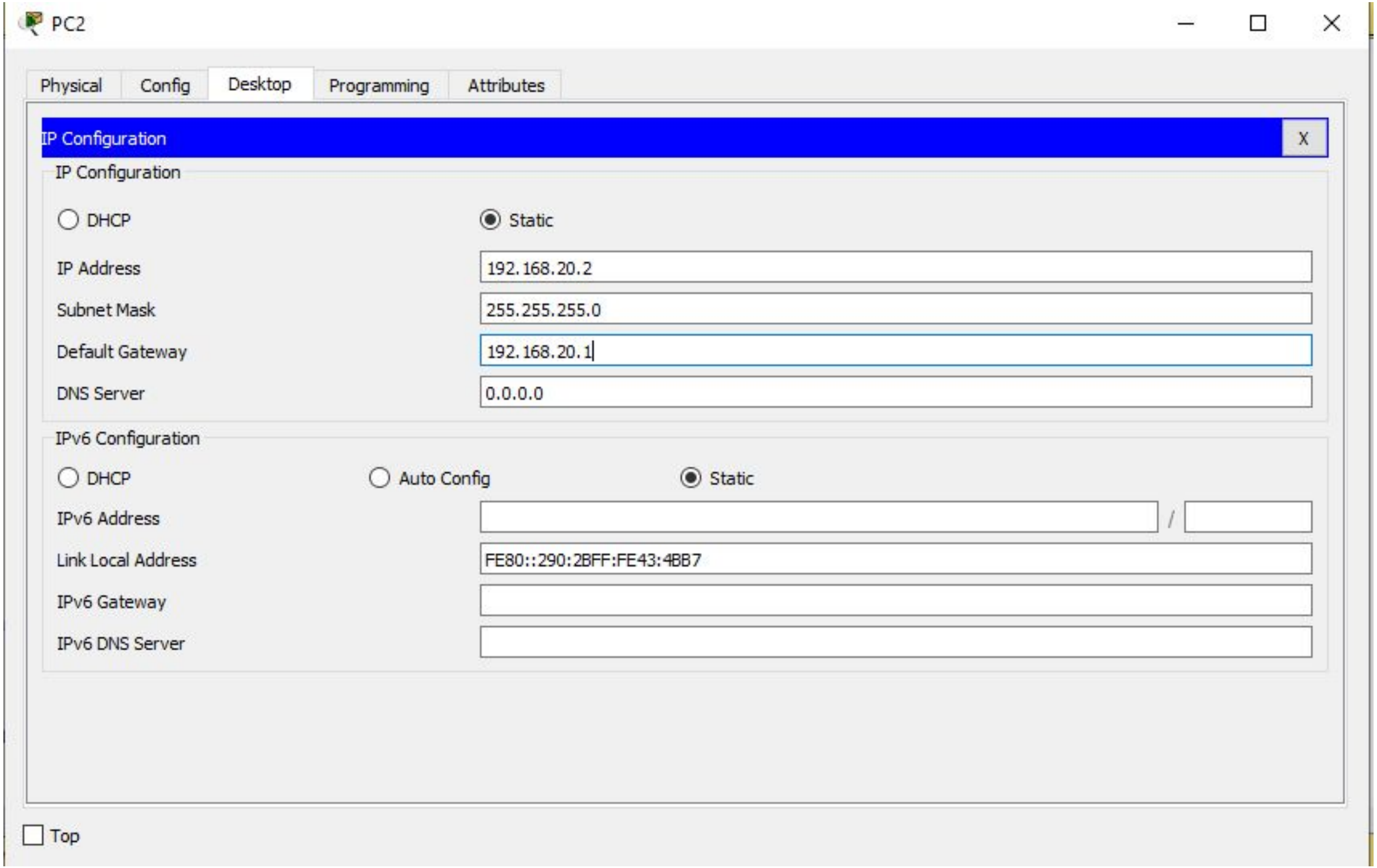
Place the devices





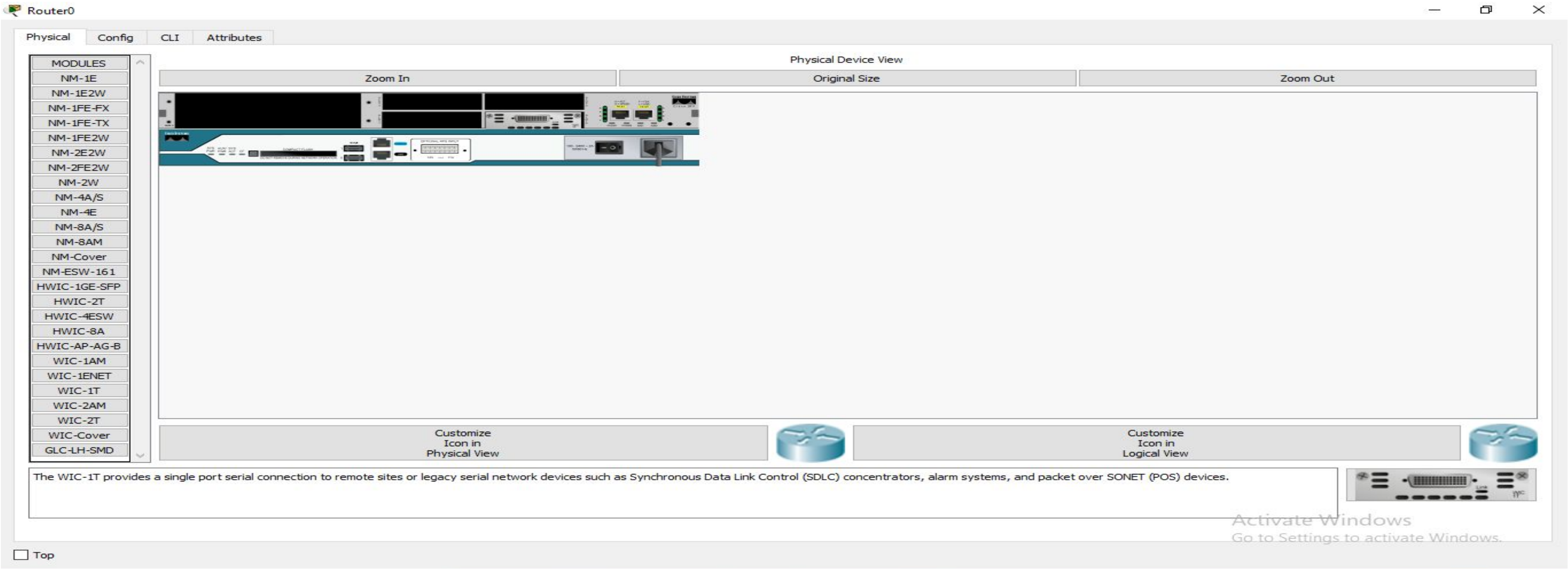
- Assign IP to all the pc and level it
- Give the gateway address all for both the LANs


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
19.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::20D:BDFF:FE84:2A0D
IPv6 Gateway
IPv6 DNS Server
Top


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::290:2BFF:FE43:4BB7
IPv6 Gateway
IPv6 DNS Server
Top

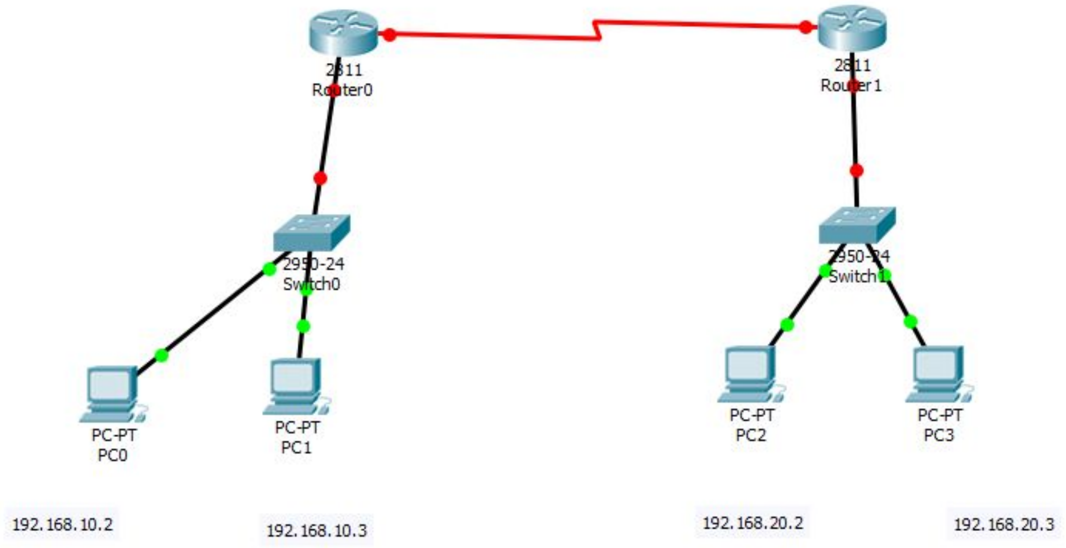
Steps

- Go the 1st router: in physical tab, switch off
- Drag and drop WIC-1T and switch on the power
- Similarly do it for the second router

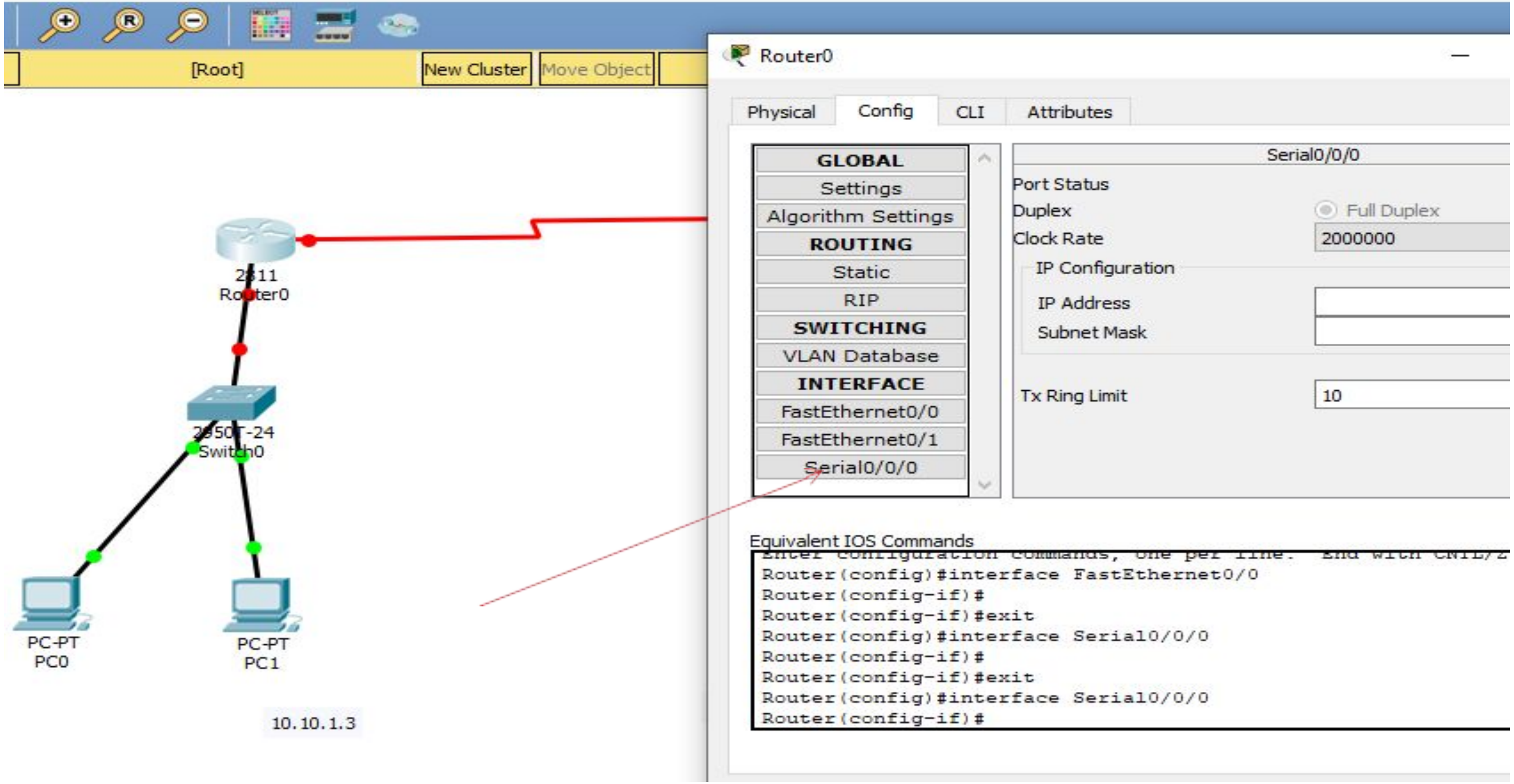


Steps

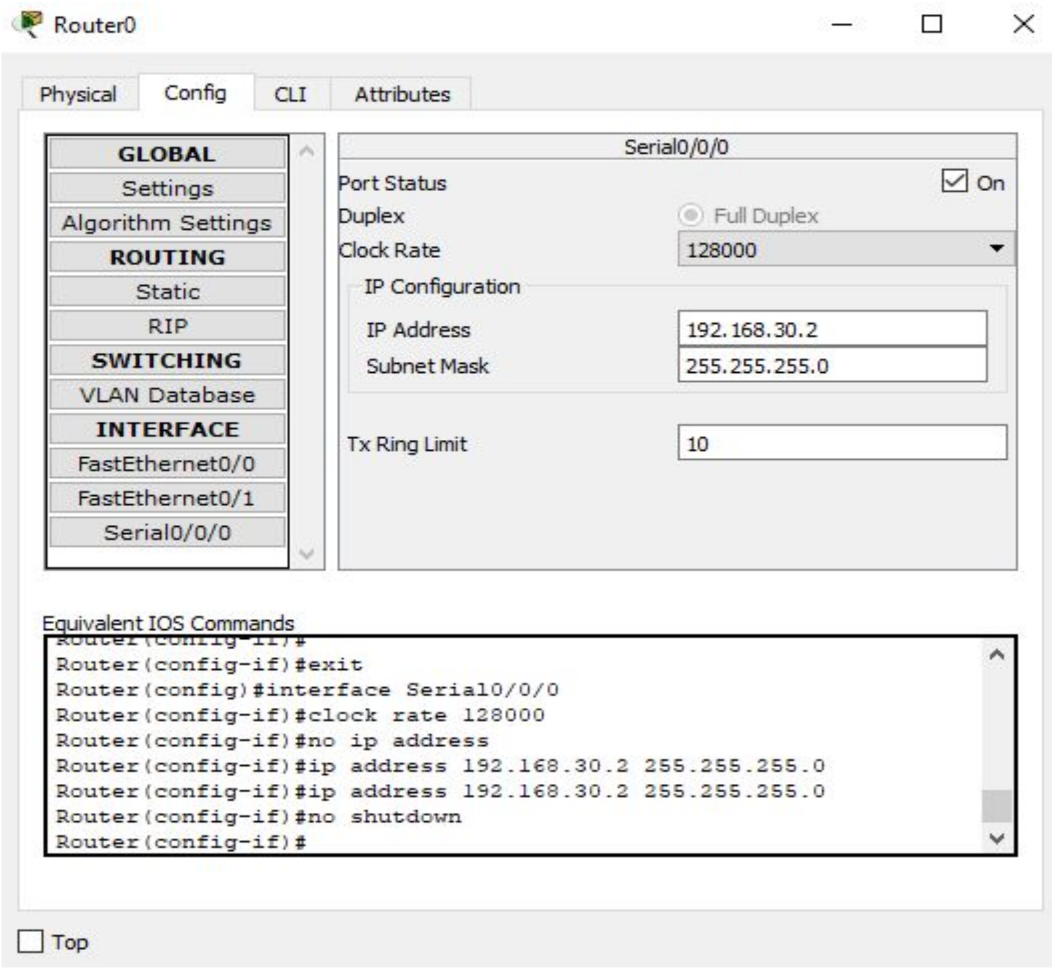
- Use Serial DCE for connection between two routers
  - Drag and drop WIC-1T and switch on the power
  - Similarly do it for the second router



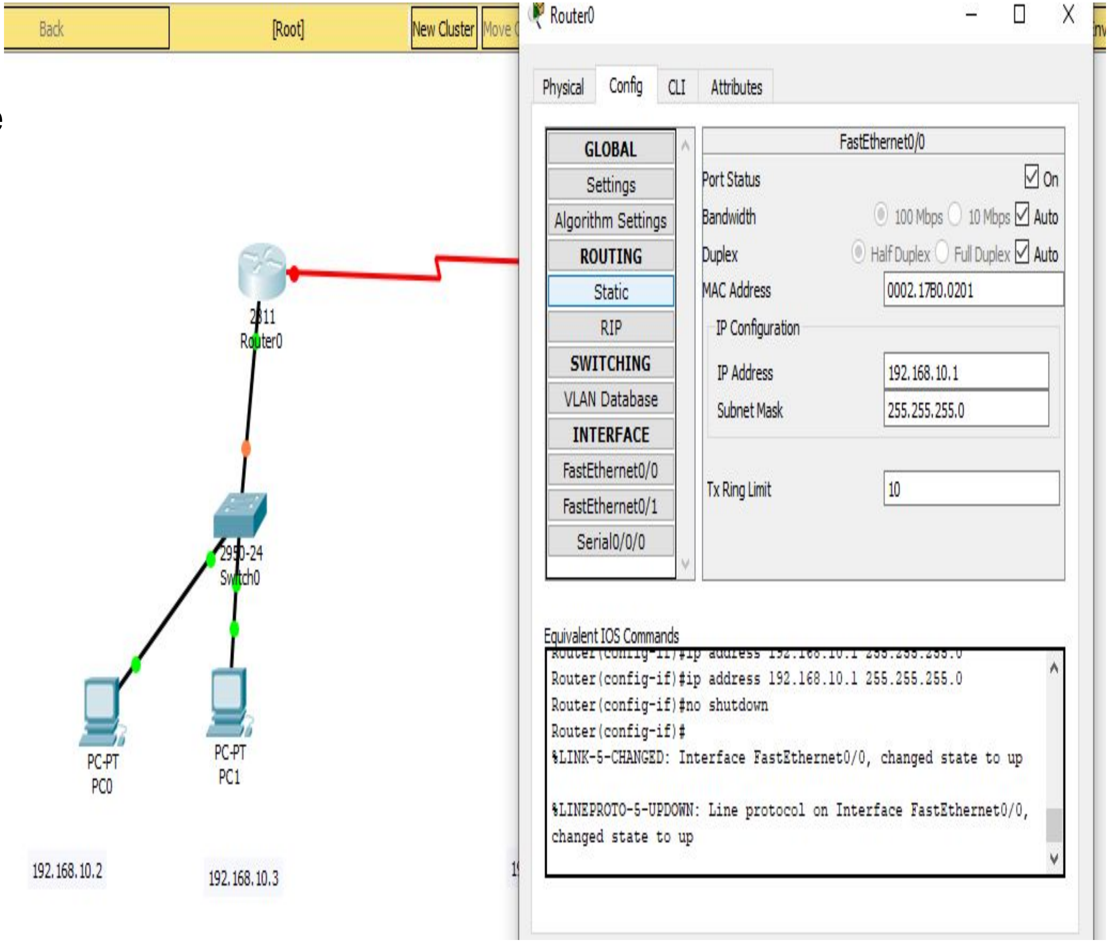
Go the router



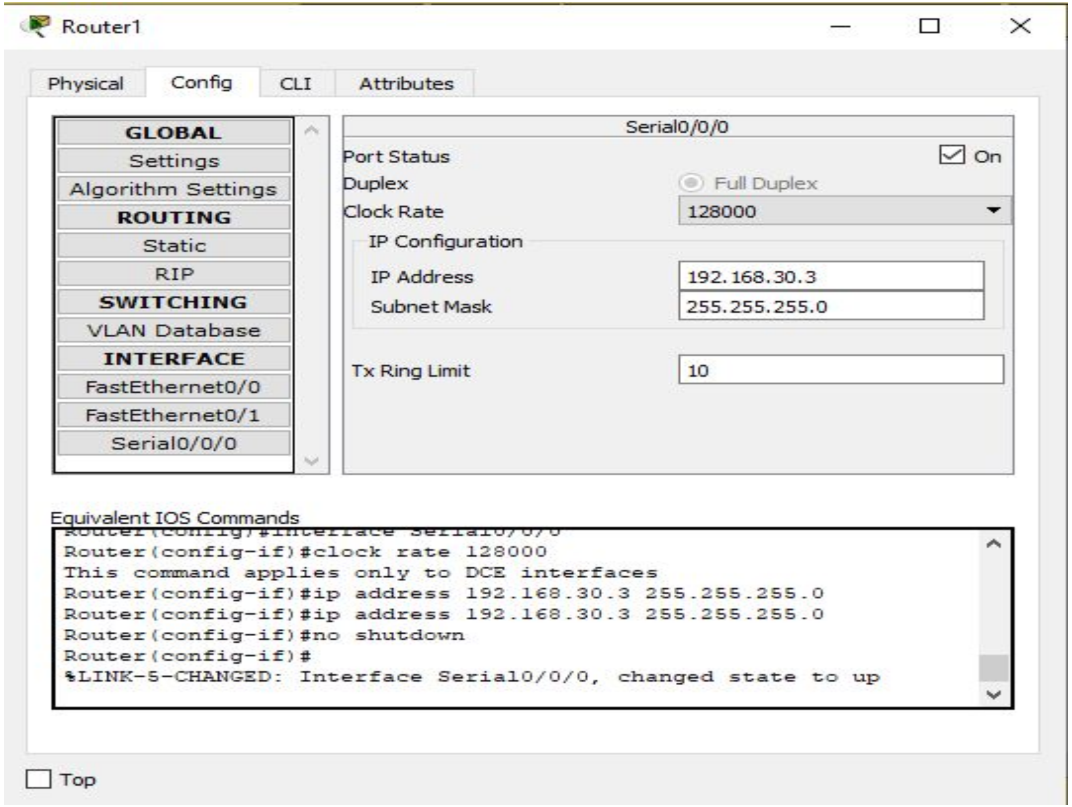
- Router > config > serial 0/0/0
- In router, clock rate is to be given 128000
- In router, give the ip
- After setting all values, set Port status ON.

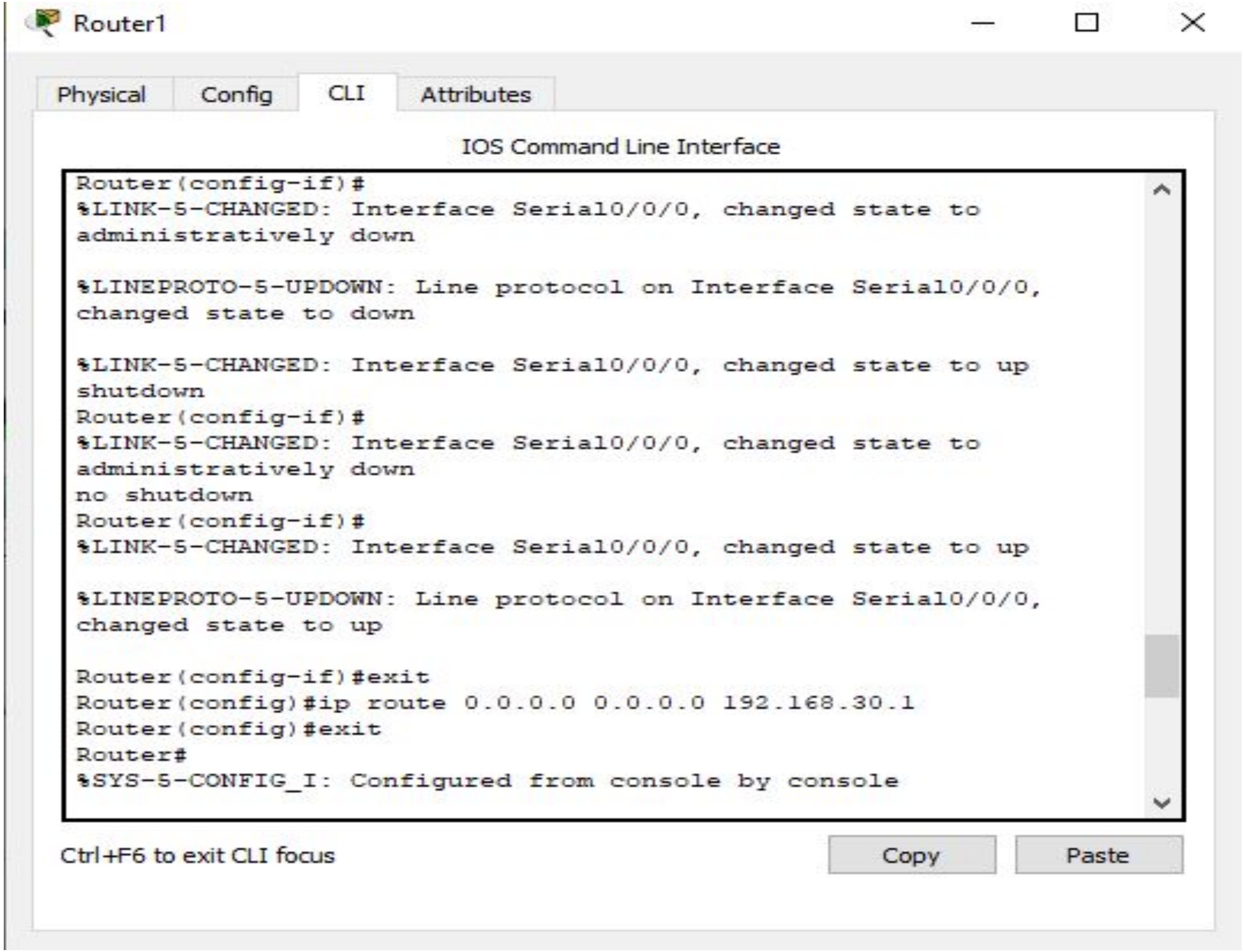


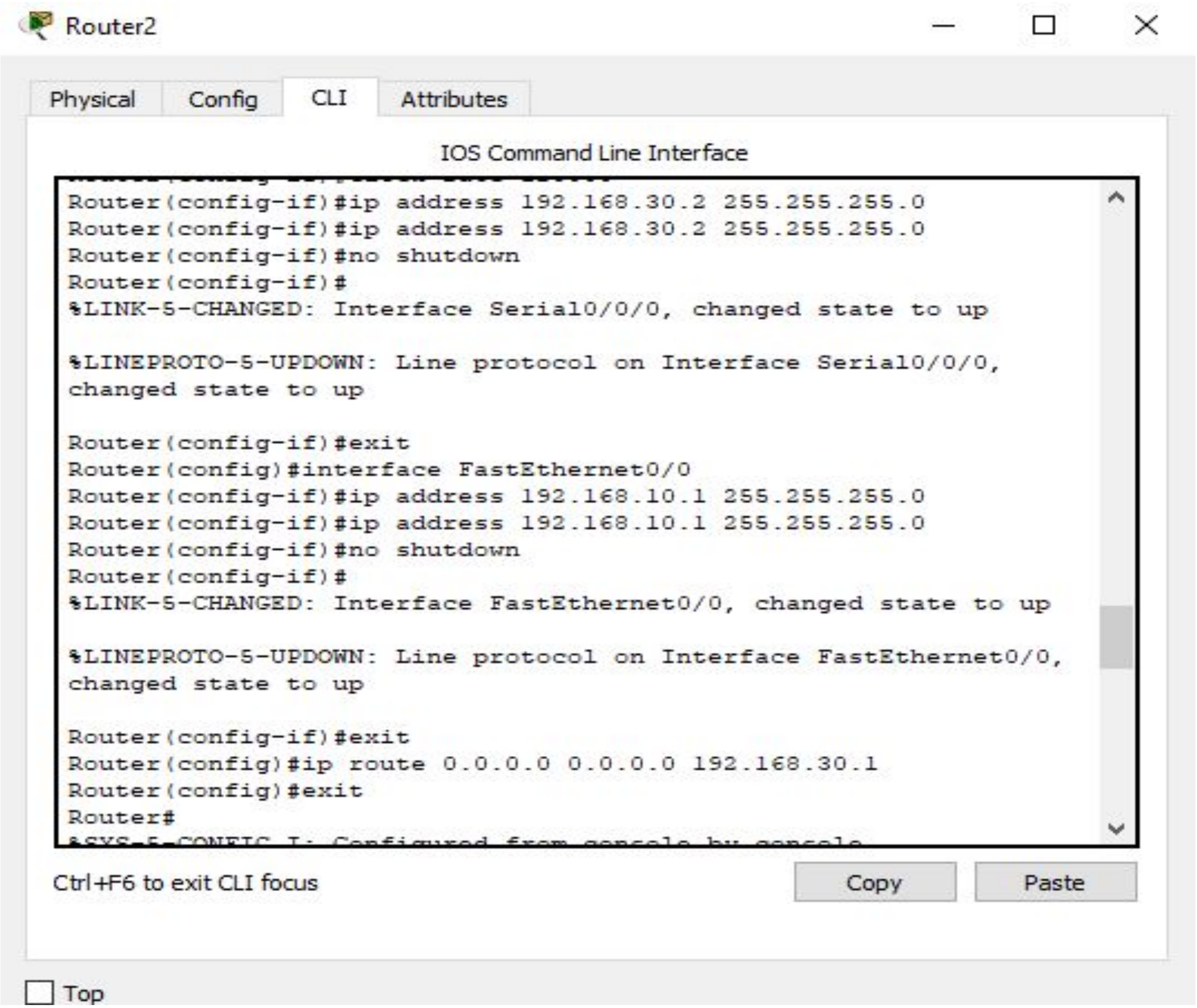
- Router > config > FastEthernet 0/0
- In router, give the IP which is same as default gateway 192.168.10.1
- Turn it on



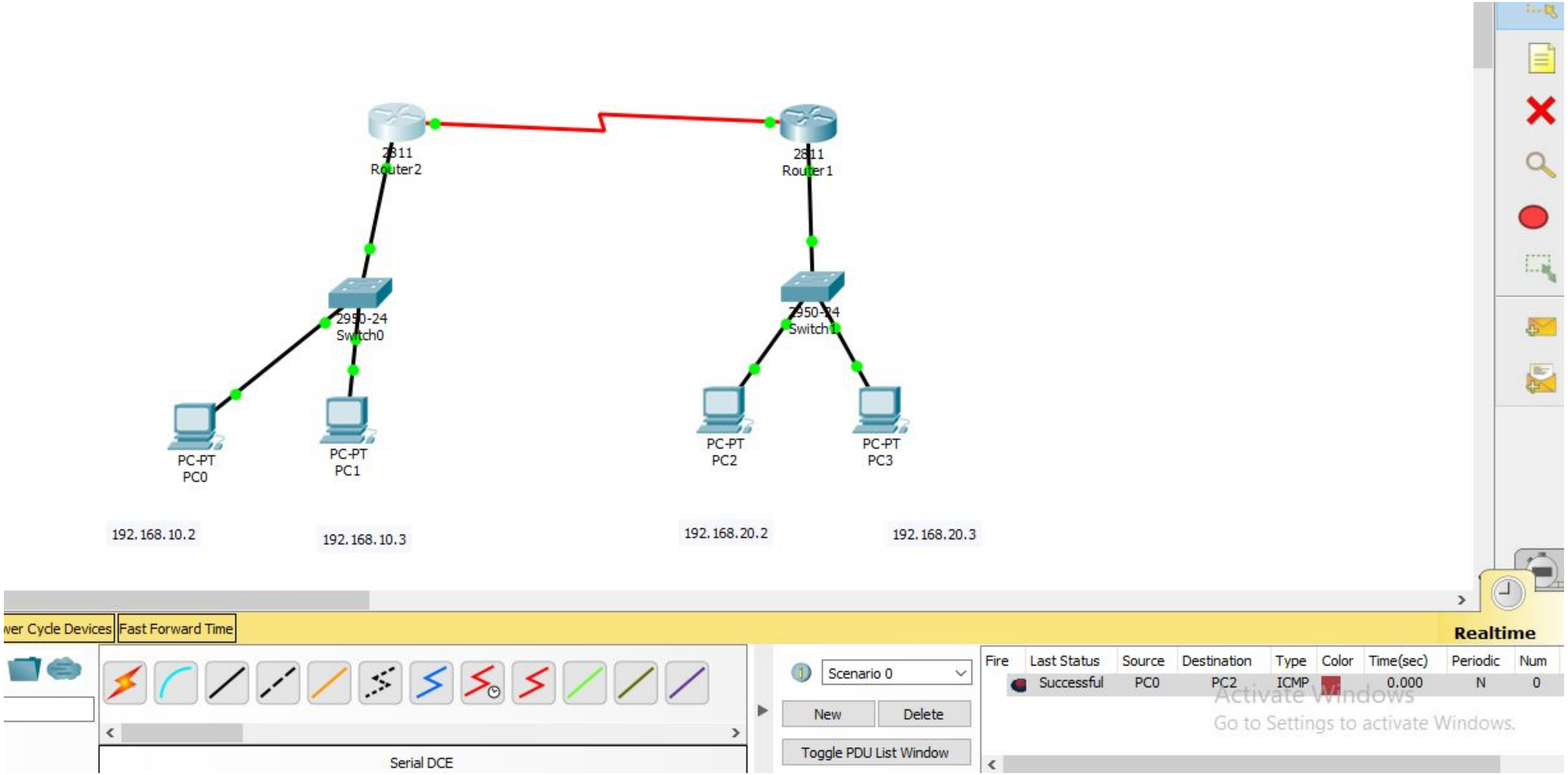
- Repeat the same procedure for the second router, Remember the Interfaces it has been connected to




Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to administratively down
no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.30.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Ctrl+F6 to exit CLI focus
Copy
Paste


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.30.1
Router(config)#exit
Router#
Ctrl+F6 to exit CLI focus
Copy
Paste
Top

Packet transmission




THANK
YOU